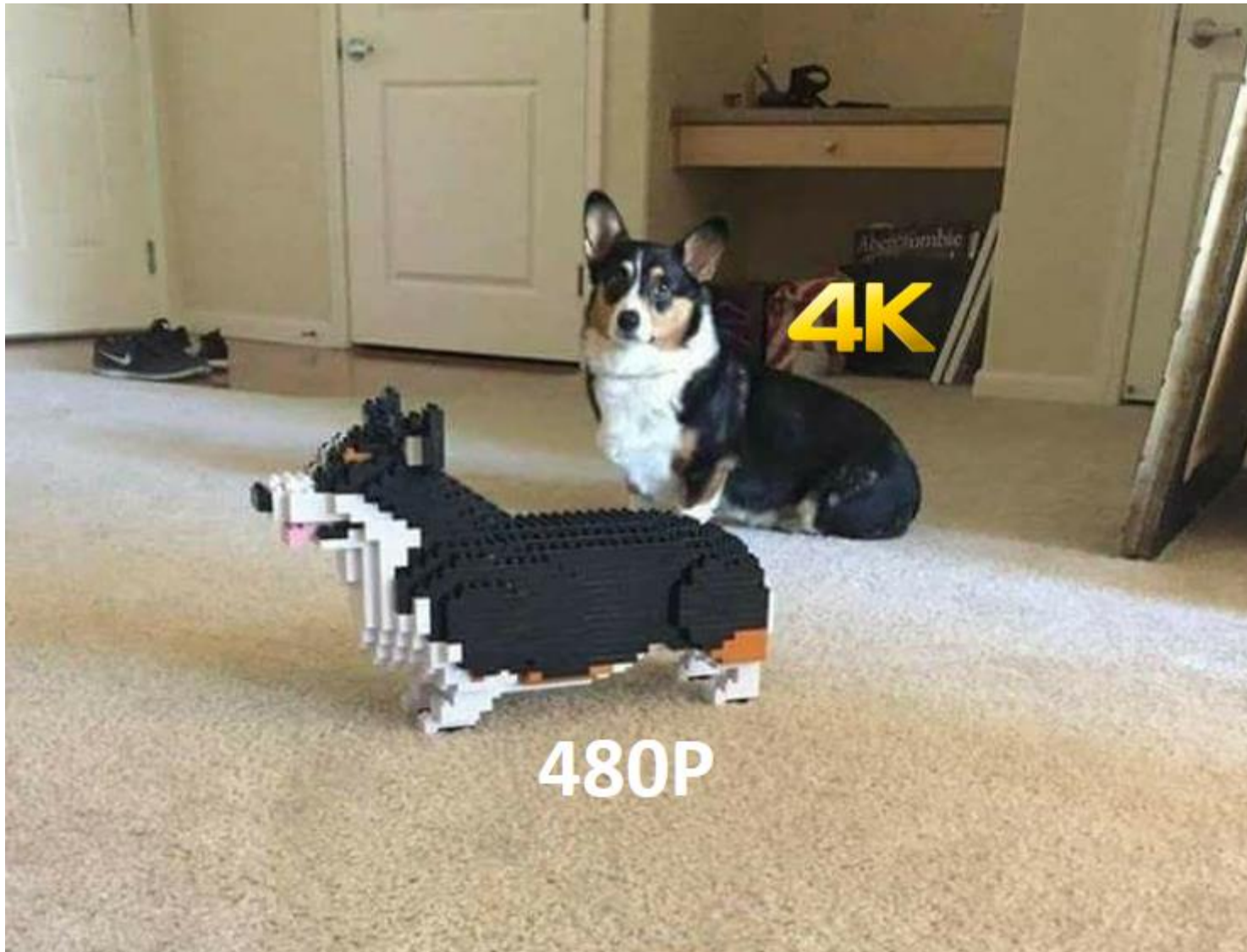




Super-Resolution

Chao-Tsung Huang

**National Tsing Hua University
Department of Electrical Engineering**



Problem definition

Downsized image



Latent image



$$Y = DBMX + n$$

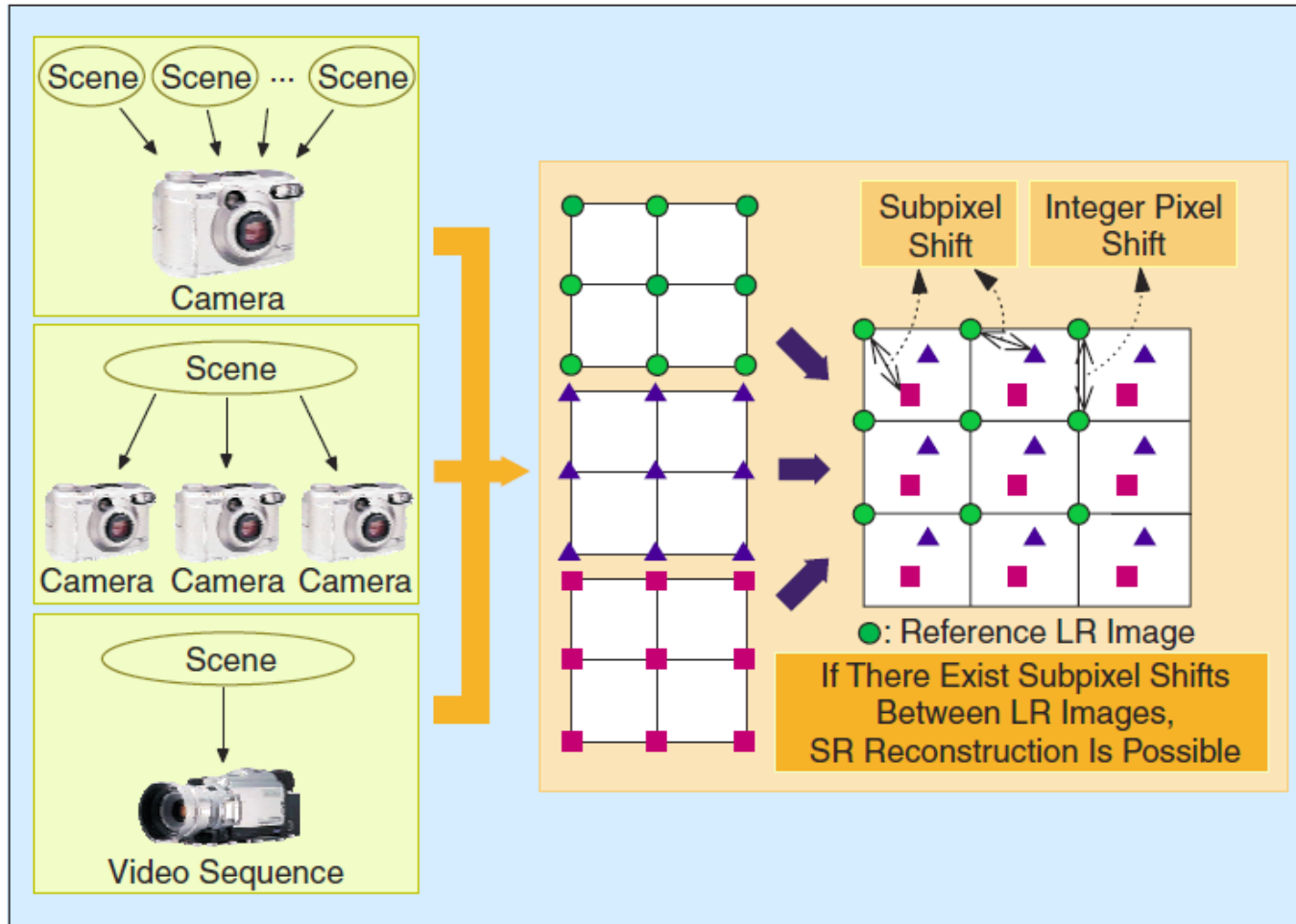
Super-resolution: Inverse problem to solve latent X from given downsized Y with model matrices: motion M , blur B , and downsampling D .



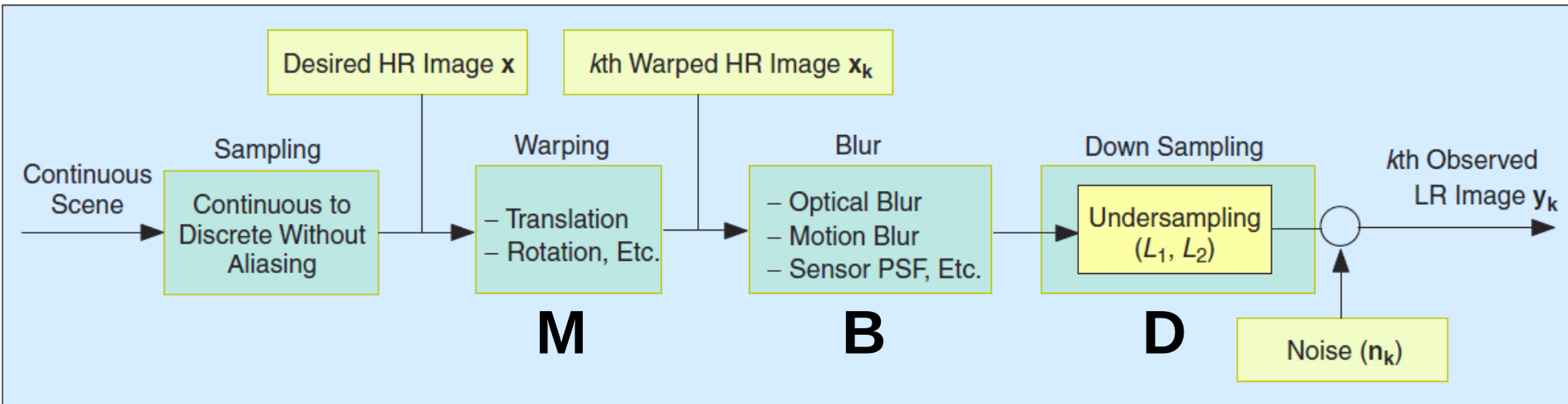
Methods to increase image size

- For small resize factor
 - Interpolation, e.g. cubic-spline → Smooth output using remained information
 - Sharpening → Enhance high-freq. components
- For large resize factor
 - **Multi-image super-resolution** → Extract aliased high-freq. information
 - **Example-based super-resolution**
└→ Learn high-freq. details from database

Multi-image super-resolution



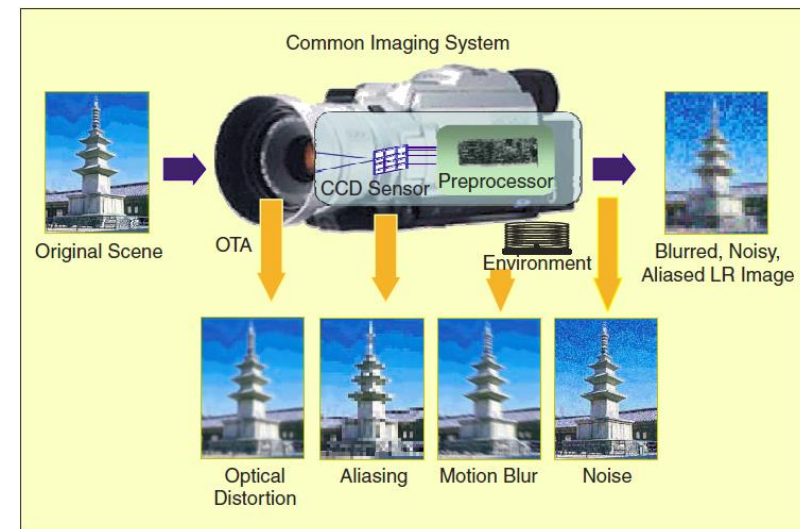
Observation model



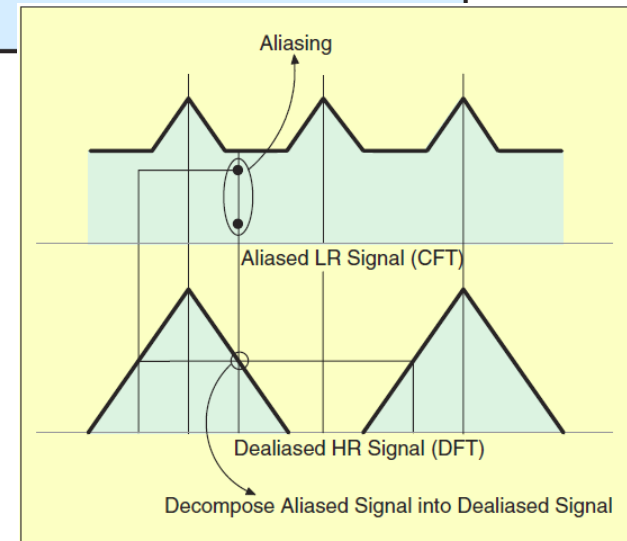
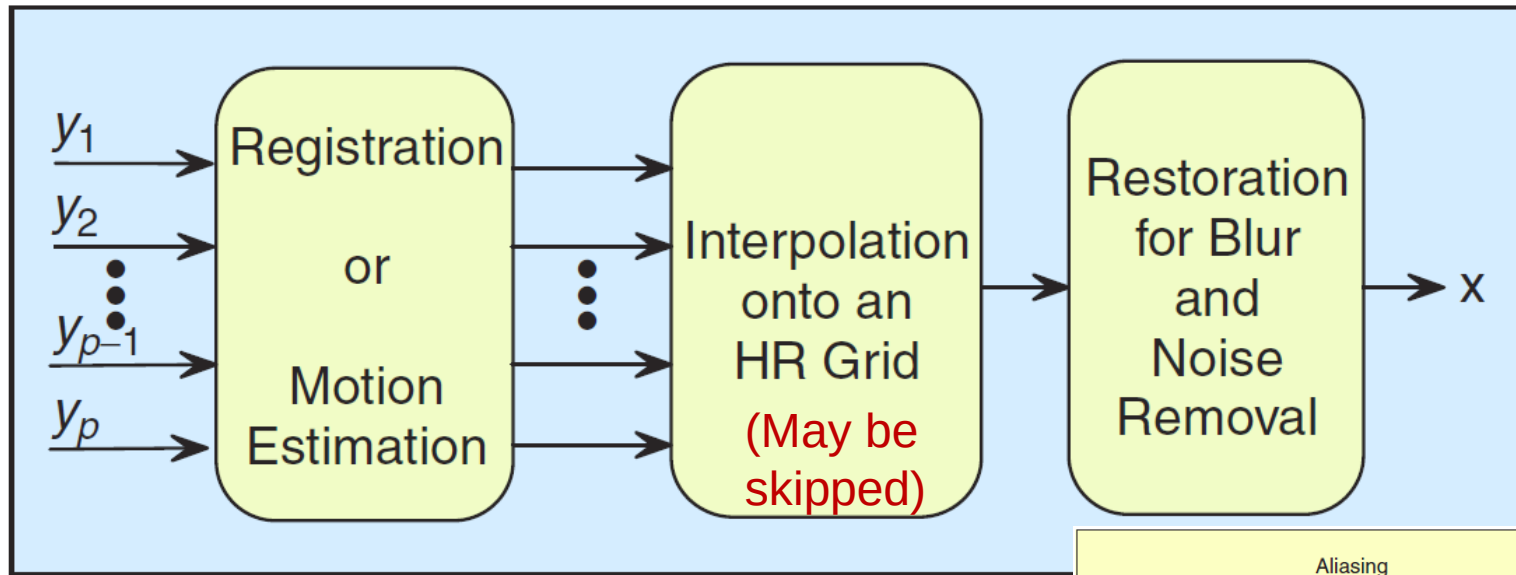
$$y_k = \mathbf{D} \mathbf{B}_k \mathbf{M}_k \mathbf{x} + \mathbf{n}_k$$



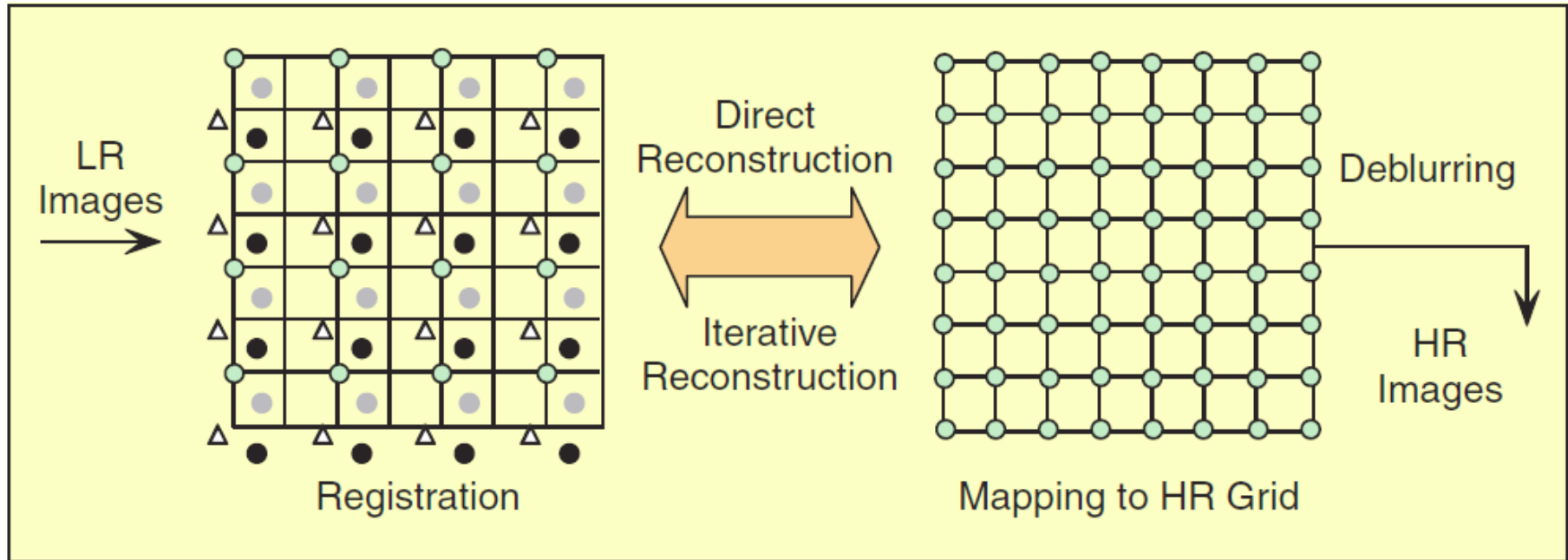
$$y_k = \mathbf{W}_k \mathbf{x} + \mathbf{n}_k$$



Scheme for super-resolution



Nonuniform interpolation approach



E.g. Iterative back-projection refines the image registration with 2D shift and rotation [1].

[M. Irani et. al., "Improving resolution by image registration," CVGIP, 1991.]

Example of 2x resizing

Low-resolution



Bi-linear
interpolation



Non-uniform
interpolation
for four LR
images



Non-uniform
interpolation +
Deblur





Optimization-based approach

Analytic solution
using least squares

$$\left[\sum_{k=1}^p \|y_k - W_k x\|^2 + \alpha \|Cx\|^2 \right]$$

Iterative solution for
edge-preserving prior

$$\hat{x} = \arg \min \left[\sum_{k=1}^p \|y - W_k \hat{x}\|^2 + \alpha \sum_{c \in S} \varphi_c(x) \right]$$

Example of 2x resizing

Low-resolution



Least squares
(small
regularization)



Least squares
(large
regularization)

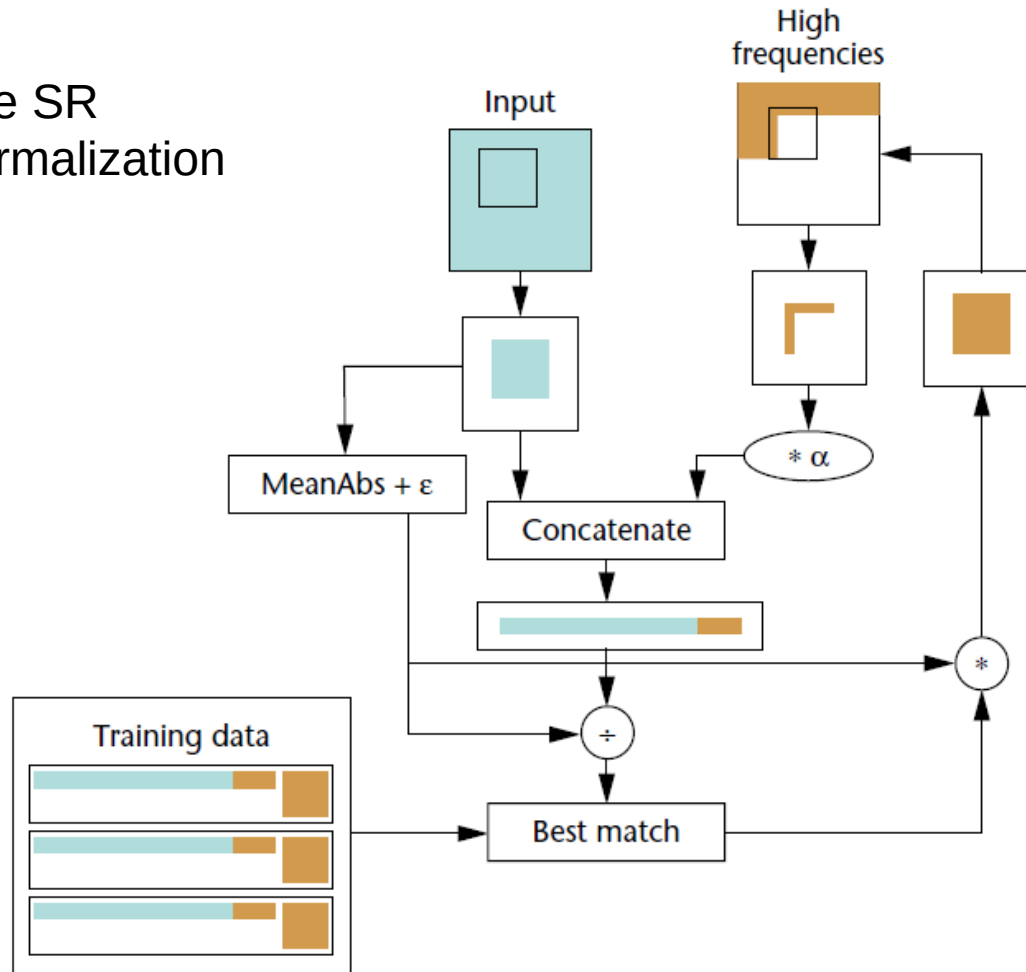


Edge-
preserving
prior



Example-based super-resolution

- Single-frame SR
- Contrast normalization



Ref : W. T. Freeman, et.al, "Example-based super-resolution," IEEE Computer Graphics and Applications, 2002.

Learn high-freq. representation from band-passed components

Low-resolution



↓ interpolation



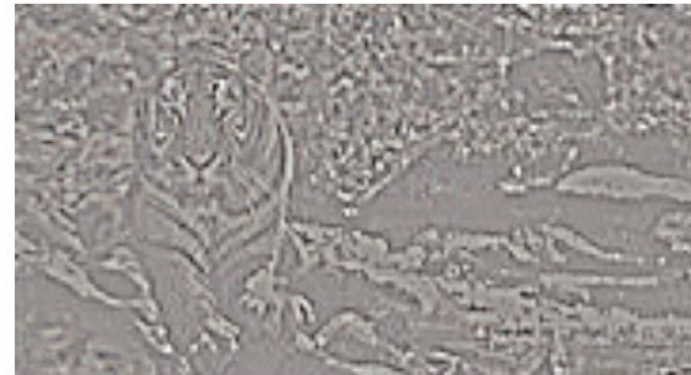
↓ Band-passed



High-resolution



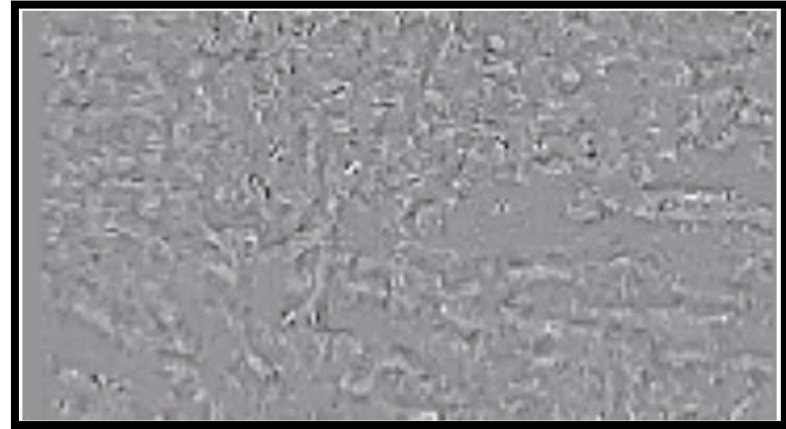
↓ High-passed



inference



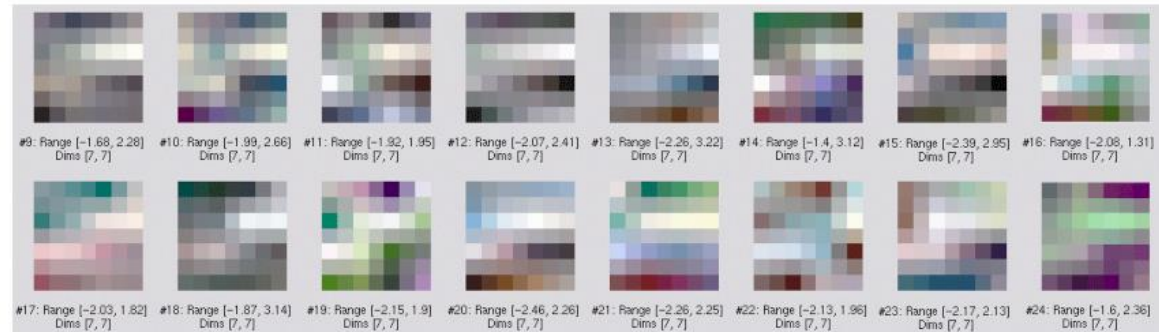
Substitute best estimation?



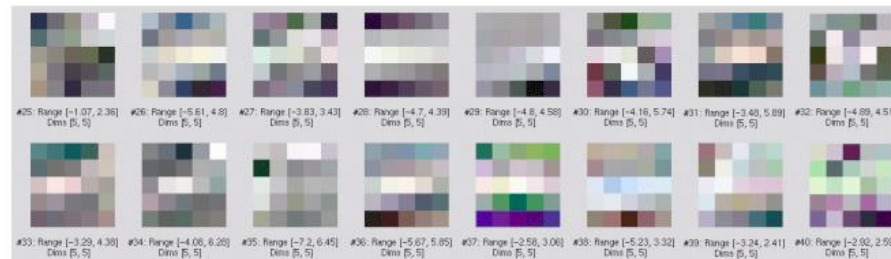
Input patch



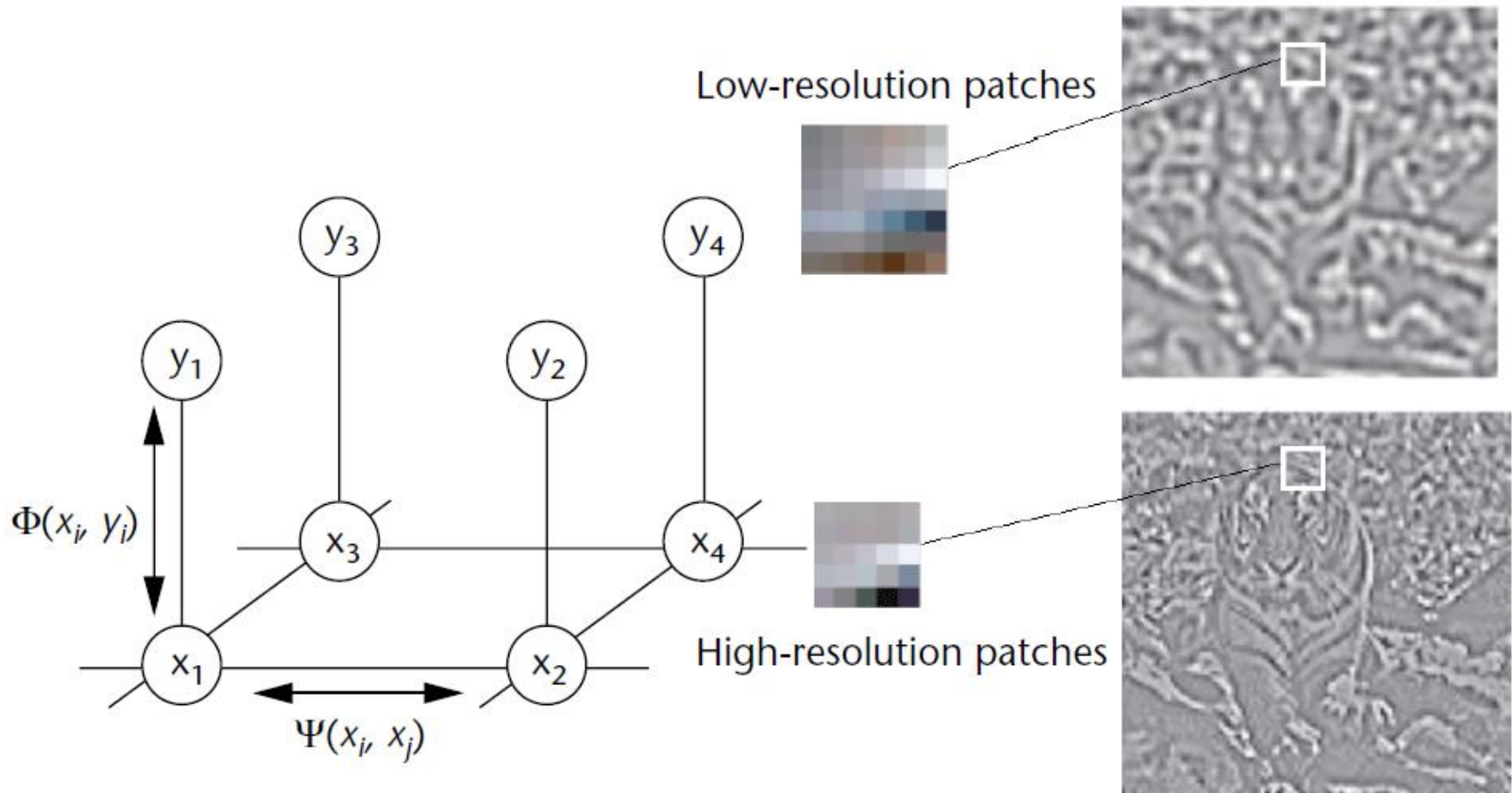
Closest image patches from database



Corresponding high-resolution patches from database



Markov network model for continuity



Result (8x)



Low-resolution



Cubic-spline interpolation



Photoshop sharpening



Example-based SR



Failure case (4x)



Original



Cubic-spline interpolation



Example-based SR

JPEG artifacts are
falsely amplified



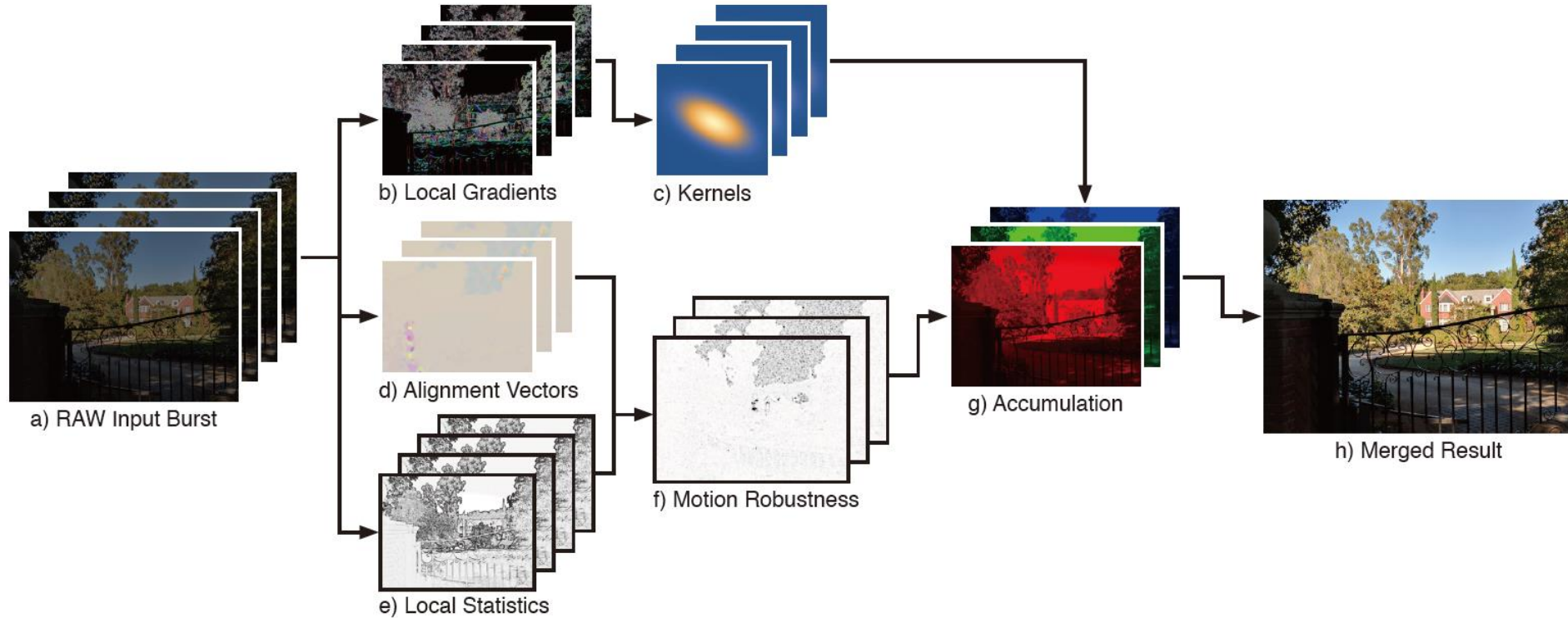
Summary

- For small resize factor
 - Interpolation, e.g. cubic-spline → Limited factor
 - Sharpening → Limited factor
- For large resize factor
 - **Multi-image super-resolution** → Requires multiple exposures and registration
 - **Example-based super-resolution** → Requires large and correct database



Appendix

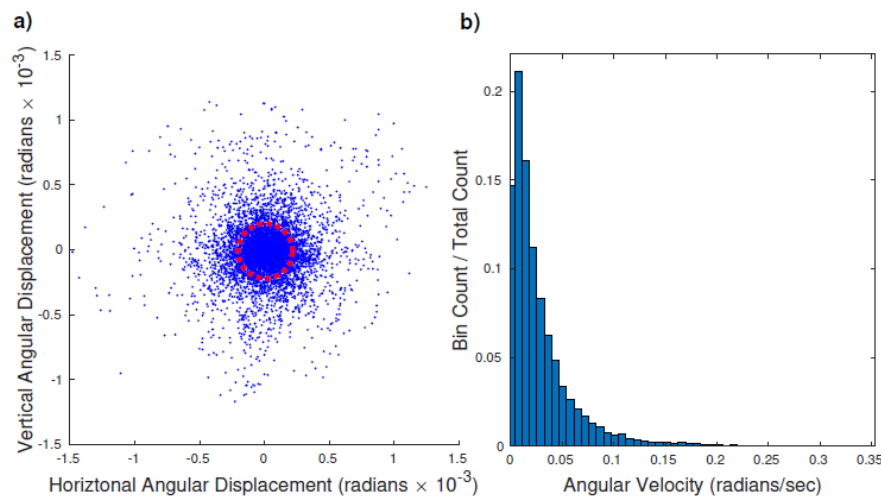
Handheld multi-frame super-resolution



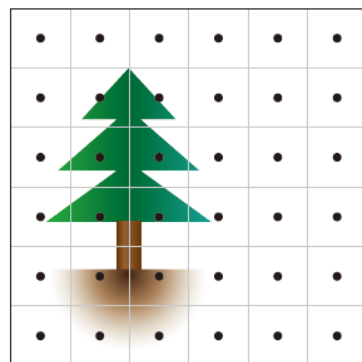
Fast and practical multi-frame super-resolution for mainstream camera phone.

[B. Wronski et. al., SIGGRAPH, 2019.]

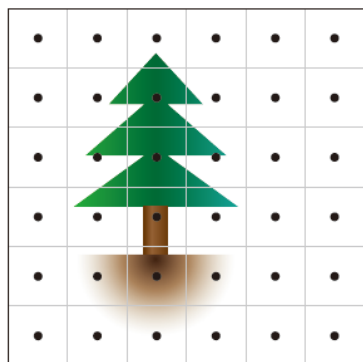
Subpixel alignment



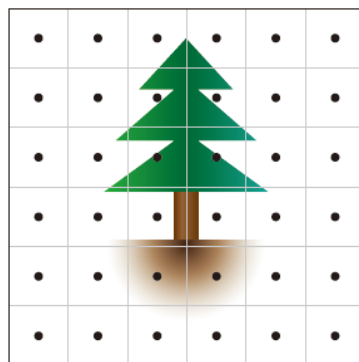
Method:
block matching then
optical flow refinement



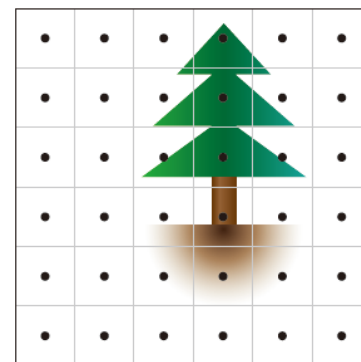
1st frame (base frame)



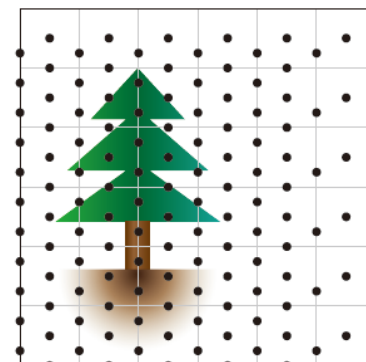
2nd frame



3rd frame

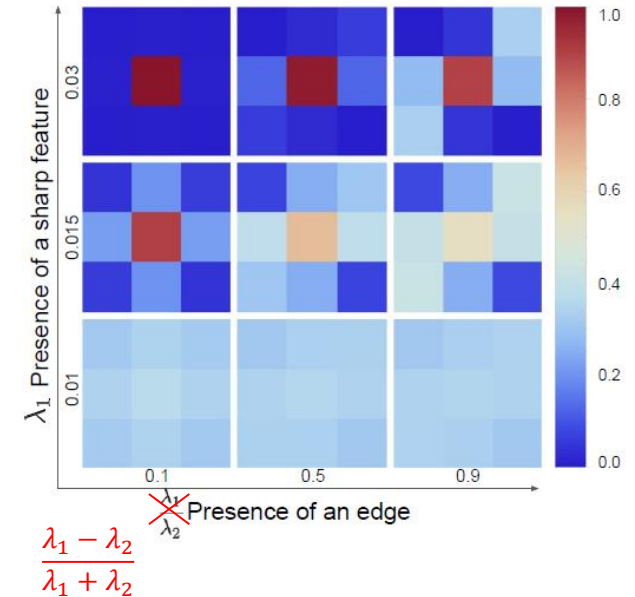
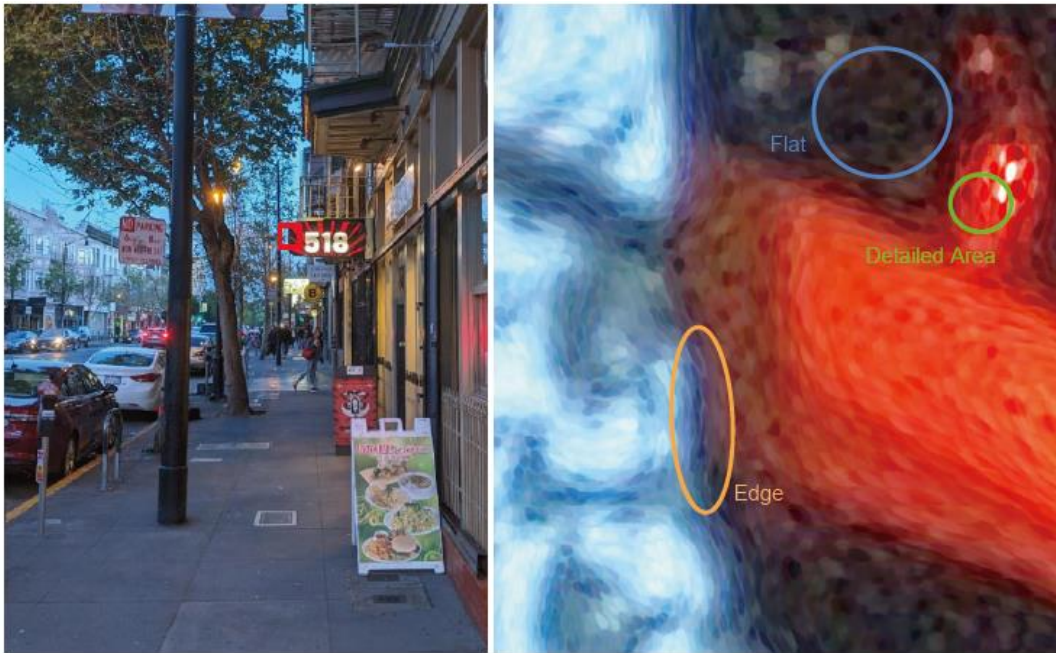


4th frame



All frames aligned to
base frame

Data reconstruction with anisotropic kernels



$$w_{n,i} = \exp \left(-\frac{1}{2} d_i^T \Omega d_i \right)$$

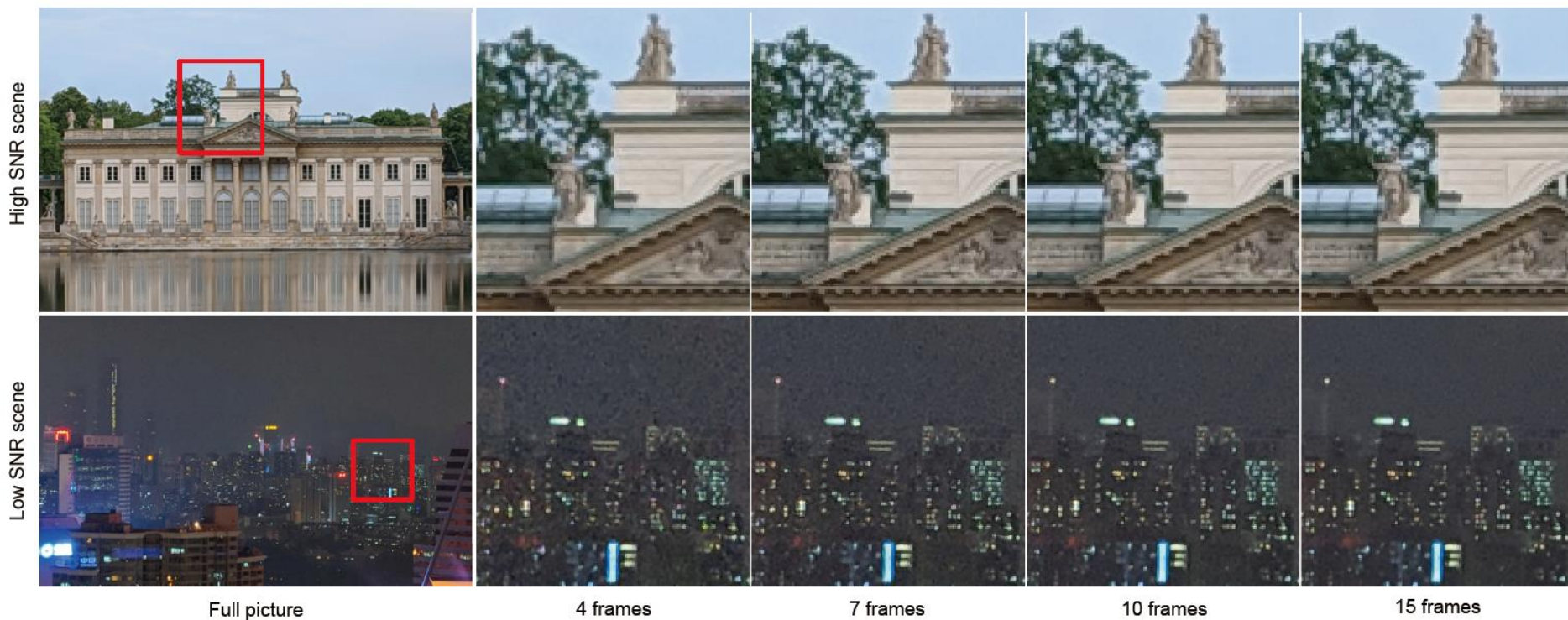
Result

Top: RGB separately and then demosaic

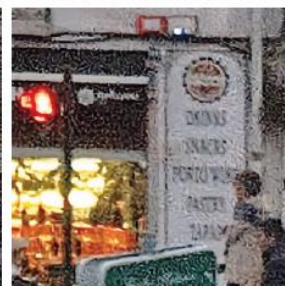
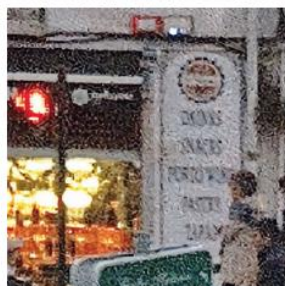
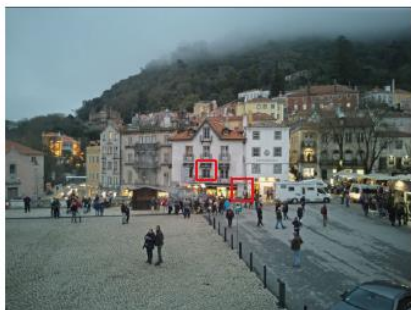


Bottom: RAW to full RGB directly

Result



Result



Full picture

VNG
+ FRVSR (blur=0.0)

DeepJoint
+ FRVSR (blur=0.0)

VNG
+ FRVSR (std=0.075)

DeepJoint
+ FRVSR (std=0.075)

Ours
+ Upscale 4x